Generation, Storage & Load Dynamics

PCEOT503: Power Systems (S5 EEE)

Bachelor of Technology (B.Tech) — 2024 Scheme

Comprehensive Study Notes & Numerical Problem Solving Suite

Prepared from Course Materials, Lecture Slides & KTU Curriculum

Module 1 Complete Syllabus Coverage:

- Conventional Sources: Hydroelectric, Thermal (Rankine, Brayton, CCPP), Nuclear
- Variable Renewables: Solar PV, Solar Thermal (CSP), Wind Turbine Systems
- Energy Storage Systems: BESS, CESS (CAES), Thermal Energy Storage (TES)
- Environmental & Ethical Factors, National & Global Regulatory Policies
- Load Curves, Load Duration Curves, Power Station Economics & 12 Numerical Problems

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Contents

1 Conventional Power Generation Technologies 2
- 1.1 Overview of Electrical Power Generation . . . 2
- 1.2 Hydroelectric Power Plants . . . 2
- 1.3 Thermal Power Plants (Steam, Gas & Combined Cycle) . . . 4
- 1.4 Nuclear Power Plants . . . 6
- 1.5 Choosing the Right Power Generation Method . . . 8

2 Variable Renewable Energy Systems 9
- 2.1 Solar Energy Systems . . . 9
- 2.2 Wind Energy Systems . . . 10
- 2.3 Renewable Energy Policies, Regulations & Future Horizons . . . 10

3 Energy Storage Systems (ESS) as Alternative Sources 12
- 3.1 Need for Energy Storage in Modern Grids . . . 12
- 3.2 Battery Energy Storage Systems (BESS) . . . 12
- 3.3 Compressed Air Energy Storage (CESS / CAES) . . . 12
- 3.4 Thermal Energy Storage (TES / Thermal SS) . . . 12
- 3.5 Comprehensive Comparison Matrix of Storage Technologies . . . 13

4 Load Curves, Station Economics & Mathematical Formulations 14
- 4.1 Load Dynamics and Operational Challenges . . . 14
- 4.2 The Load Curve and Load Duration Curve . . . 14
- 4.3 Mathematical Definitions of Economic Factors . . . 14
- 4.4 Fully Worked-Out Numerical Problems . . . 15

5 Quick Revision Sheet & Model Examination Questions 19
- 5.1 Master Formula and Definition Summary Table . . . 19
- 5.2 Master Comparison Matrix of All Power Generation Sources . . . 19
- 5.3 Model Examination Questions (KTU S5 Pattern) . . . 20

1 Conventional Power Generation Technologies

Syllabus Requirement: Generation from renewable and non-renewable sources – Hydro, thermal, nuclear – (block schematic details, environmental and ethical factors, advantages, disadvantages)

1.1 Overview of Electrical Power Generation

Electrical energy is a secondary energy carrier derived from primary resources found in nature. The global power system has historically relied on three proven conventional generation technologies:

1. **Thermal Power Plants:** Convert the chemical energy of fossil fuels into heat, generating high-pressure, superheated steam that drives rotating steam turbines.
2. **Hydroelectric Power Plants:** Convert the gravitational potential energy of water stored at elevated heads into mechanical and electrical power.
3. **Nuclear Power Plants:** Exploit controlled nuclear fission chain reactions to release binding energy from atomic nuclei to generate steam.

These conventional generation assets form the **baseload bedrock** of modern synchronous power grids due to their high availability, dispatchability, and inherent synchronous rotational inertia.

1.2 Hydroelectric Power Plants

Working Principle & Energy Conversion

A hydroelectric power plant utilizes the potential energy of water stored in an elevated reservoir. As water flows through closed pressurized conduits called **penstocks**, potential energy converts to kinetic energy. The resulting high-velocity water jet strikes the runner blades of a **hydraulic turbine**, converting hydraulic energy into rotational mechanical energy. An electromechanically coupled **synchronous generator** converts this mechanical torque into 3-phase AC electricity via electromagnetic induction.

Energy Conversion Chain in Hydroelectric Plants:

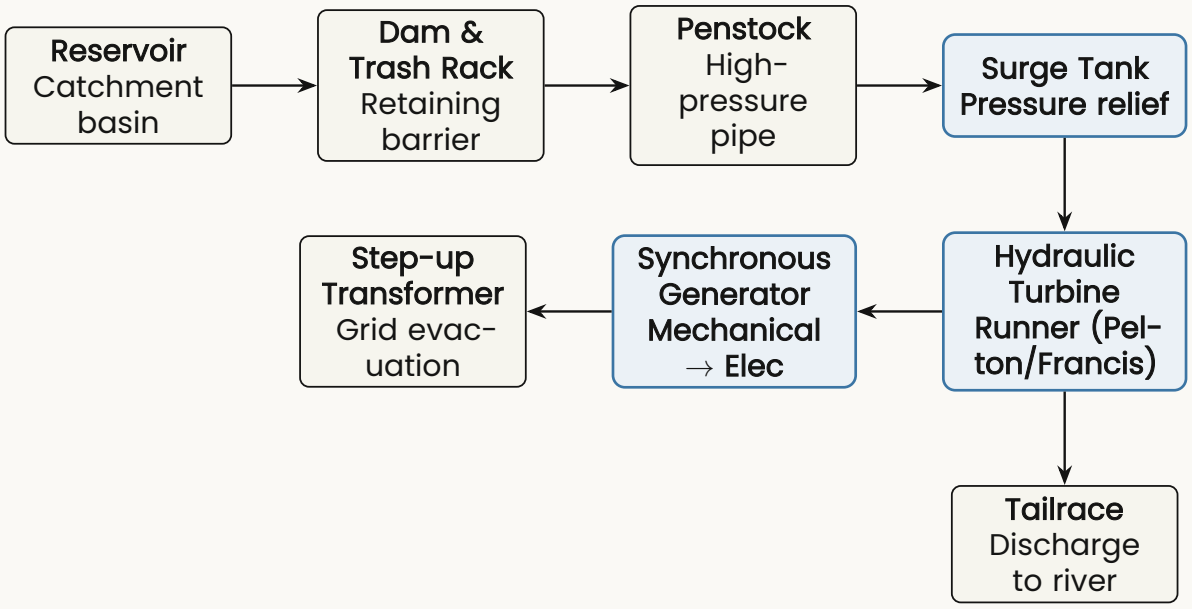
$$\text{Gravitational Potential Energy} \xrightarrow{\text{Penstock}} \text{Kinetic Energy} \xrightarrow{\text{Hydraulic Turbine}} \text{Mechanical Energy} \xrightarrow{\text{Generator}} \text{Electrical Energy}$$

The theoretical hydraulic power developed at the turbine shaft is given by:

$$P = \rho \cdot g \cdot Q \cdot H \cdot \eta \quad \text{[Watts]} \tag{1}$$

where $\rho = 1000\,\text{kg/m}^3$ (water density), $g = 9.81\,\text{m/s}^2$ (acceleration due to gravity), Q is discharge volume rate (m^3/s), H is the effective net head (m), and η is overall plant efficiency.

Block Schematic & Major Plant Components



1. **Storage Reservoir:** Stores river water during high-inflow monsoon seasons to guarantee year-round operational supply.
2. **Dam:** A civil engineering barrier (gravity, arch, or earth-rockfill) built across a river to create the required hydrostatic head.
3. **Forebay & Trash Racks:** Located at the penstock intake; screens prevent floating debris, tree branches, and silt from entering the pressurized pipe.
4. **Penstock:** Heavy-walled steel or prestressed concrete conduits conveying high-pressure water from reservoir to turbine.
5. **Surge Tank:** A vertical standing pipe or open shaft located on long penstocks near the powerhouse. It mitigates dangerous hydraulic pressure surges during sudden governor valve closures (*water hammer effect*) and supplies supplemental water during sudden load pick-ups.
6. **Hydraulic Turbine:** Extracts energy from moving water:
   - *Impulse Turbine (Pelton Wheel):* High head (> 300 m), low discharge rate.
   - *Reaction Turbine (Francis Turbine):* Medium head ($30 - 300$ m), medium discharge rate.
   - *Axial Flow Turbine (Kaplan Turbine):* Low head (< 30 m), high discharge volume.
7. **Draft Tube:** An airtight diverging discharge pipe fitted to reaction turbines to recover kinetic exit energy.
8. **Tailrace:** The exit channel discharging spent water back into the natural downstream river.

Key Operational Features & Real-World Benchmarks

- **Rapid Start-up & Load Following:** Hydro units can ramp from complete standstill to 100% rated power within 2 to 5 minutes, making them prime assets for **peak-load dispatch, spinning reserve, and grid black start**.
- **High Velocity Jets:** Water falling through large hydroelectric dams can exceed speeds of 100 **km/h** before striking the turbine runner.
- **Idukki Hydroelectric Project (Kerala):** Installed capacity 780 MW (6×130 MW Pelton units). Built across the Periyar River between the two granite hills *Kuravan* and *Kurathi*. It is India's first and one of Asia's largest double-curvature parabolic **arch dams** (168.9 m height), feeding an underground powerhouse at Moolamattom. During monsoon peaks, opening the spillway gates attracts thousands of visitors.

- **Tehri Dam (Uttarakhand):** India's tallest dam standing at 260.5 m (similar in height to the Eiffel Tower), supplying $1,000$ MW of peaking power.
- **Three Gorges Dam (China):** The world's largest power station ($22,500$ MW). In 2005, NASA scientist **Dr. Benjamin Fong Chao** calculated that when full (39.3×10^9 m^3 of water mass), the Three Gorges reservoir shifted Earth's mass distribution enough to increase the length of a day by approximately 0.06 μs.

Advantages & Disadvantages of Hydroelectric Power

Advantages: Clean and renewable with zero operational greenhouse gas emissions; negligible operating costs (no fuel); long operating life ($> 80 - 100$ years); fast response for peak load and frequency control; provides multipurpose flood control and irrigation benefits.
Disadvantages: Very high upfront capital expenditure; long construction gestation periods (often $6 - 10$ years); strict dependence on seasonal monsoon rains; geographically constrained to hilly terrain distant from load centers.

Environmental & Ethical Considerations

- **Environmental:** Inundation of vast forest and wildlife habitats; physical barrier to fish migration (partially mitigated by artificial *fish ladders* or *fish elevators*); anaerobic decay of submerged vegetation generating biogenic methane (CH_4); altered downstream sediment transport.
- **Ethical & Social:** Involuntary resettlement of indigenous and rural populations; equitable compensation and livelihood rehabilitation; interstate and international river water-sharing disputes (e.g., Cauvery basin); submergence of cultural and historical heritage.

1.3 Thermal Power Plants (Steam, Gas & Combined Cycle)

Working Principle & Thermodynamic Cycles

A thermal power plant converts the chemical energy of fuel into thermal energy to generate high-pressure, superheated steam that drives a steam turbine.

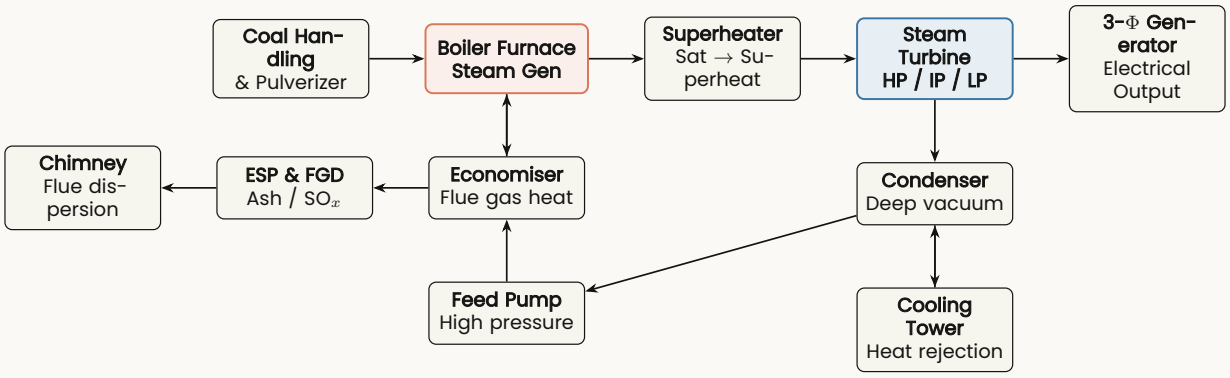
Energy Conversion Chain in Thermal Plants:

$$\text{Chemical Energy (Fuel)} \xrightarrow{\text{Combustion}} \text{Thermal Energy} \xrightarrow{\text{Steam Turbine}} \text{Mechanical Energy} \xrightarrow{\text{Generator}} \text{Electrical Energy}$$

Thermodynamic Classifications

1. **Steam Power Plant (Rankine Cycle):** Water/steam circulates in a closed loop undergoing pressurized heating, evaporation, multi-stage turbine expansion, condensation, and pumping.
2. **Gas Turbine Power Plant (Brayton Cycle):** Compressed air is mixed with natural gas or diesel in a combustor, and expanding hot combustion gases directly rotate a gas turbine ($10 - 15$ min quick start for peaking duty).
3. **Combined Cycle Power Plant (CCPP):** Integrates a gas turbine (*Brayton topping cycle*) with a steam turbine (*Rankine bottoming cycle*). Hot exhaust gases ($> 500^\circ$C) pass through a **Heat Recovery Steam Generator (HRSG)** to generate steam for a second turbine, achieving overall efficiencies **exceeding 60%** (compared to $35 - 42\%$ for standalone steam cycles).

Block Schematic of a Coal-Fired Thermal Power Plant



Major Sub-systems and Auxiliary Functions

- **Boiler & Superheater:** Pulverized coal combustion evaporates water. Saturated steam is superheated to $540 - 600°C$ to prevent moisture condensation and blade erosion inside the low-pressure turbine stages.
- **Economiser & Air Preheater (APH):** Recover heat from exiting flue gas. The economiser preheats boiler feedwater, and the APH warms combustion air, increasing boiler efficiency by $5 - 10\%$.
- **Condenser:** Cools exhaust steam back to liquid water while maintaining a **high vacuum** ($0.05 - 0.08$ bar absolute), maximizing enthalpy drop and thermal efficiency.
- **Cooling Tower:** Dissipates heat from circulating condenser water. The white plume seen leaving cooling towers is **pure condensed water vapor**, not smoke.
- **Electrostatic Precipitator (ESP):** Applies high-voltage DC electric fields ($40 - 60$ kV) to capture over 99.5% of particulate fly ash before flue gas enters the stack.
- **Centrifugal Governor Valves:** Regulate steam mass flow into the high-pressure turbine to maintain constant synchronous speed (3000 RPM for 50 Hz, 2-pole) under changing electrical loads.

Fuels Utilized in Thermal Stations

- **Coal:** Mainstay for large baseload plants (Anthracite, Bituminous, Lignite).
- **Natural Gas:** Clean-burning methane fuel with lower carbon emissions per kWh.
- **Naphtha:** Liquid hydrocarbon mixture derived from crude refining. Used in plants such as *NTPC Rajiv Gandhi CCPP (Kayamkulam, Kerala, 359 MW)*, which ran on naphtha before switching to pipeline natural gas via the Kochi–Mangaluru pipeline.
- **Low Sulphur Heavy Stock (LSHS) / Diesel:** Heavy residual fuel used for normal operation at *Brahmapuram Diesel Power Plant (106.6 MW, Kochi)* and *Kozhikode (Nallalam) Diesel Power Project (128 MW)*, which use diesel for starting and LSHS during normal operation.
- **Biomass:** Agricultural residues and wood pellets co-fired with coal to lower net carbon intensity.

Thermal Engineering Benchmarks & USC Technology

- Modern steam turbines operate at temperatures of $540 - 600°C$ and pressures exceeding 25 **MPa**.
- A single 1000 MW thermal station can supply electricity to approximately $700,000 - 1,000,000$ **homes**.
- **Ultra-Supercritical (USC)** plants operate above the critical point of water (22.064 MPa, $373.95°C$), reaching efficiencies of **up to 47%** and cutting emissions per MWh by $15 - 20\%$.

- **Prominent Thermal Plants:** *Vindhyachal Super Thermal Power Station (Madhya Pradesh, 4,760 MW)* — India's largest thermal plant; *Taichung Power Plant (Taiwan, 5,500 MW)* — one of the world's largest coal stations.

Advantages & Disadvantages of Thermal Power

Advantages: Highly reliable, continuous 24/7 baseload supply; lower upfront capital cost per MW and shorter gestation time than nuclear/hydro; flexible siting near coal mines (pit-head) or load centers.
Disadvantages: Heavy greenhouse gas emissions (CO_2) and air pollutants (SO_2, NO_x, PM2.5); depletes fossil fuel reserves; requires substantial cooling water; creates fly ash disposal challenges.

Environmental & Ethical Considerations

- **Environmental:** Major contributor to climate change (CO_2); acid rain (SO_2); ground-level smog (NO_x); heavy metals (arsenic, lead, mercury) in fly ash ponds; thermal water discharge pollution.
- **Ethical:** Balancing affordable energy access against chronic public health issues in neighboring communities; ensuring a **Just Transition** for coal-dependent workforces as economies decarbonize.

1.4 Nuclear Power Plants

Working Principle & Nuclear Fission Physics

Nuclear power plants release the binding energy of atomic nuclei via controlled nuclear fission. When a fissile **Uranium-235 (^{235}U)** nucleus absorbs a thermal neutron, it splits into lighter fission fragments, releasing 2 to 3 fast neutrons and approximately 200 MeV of energy per event.

Nuclear Fission Reaction Formula:

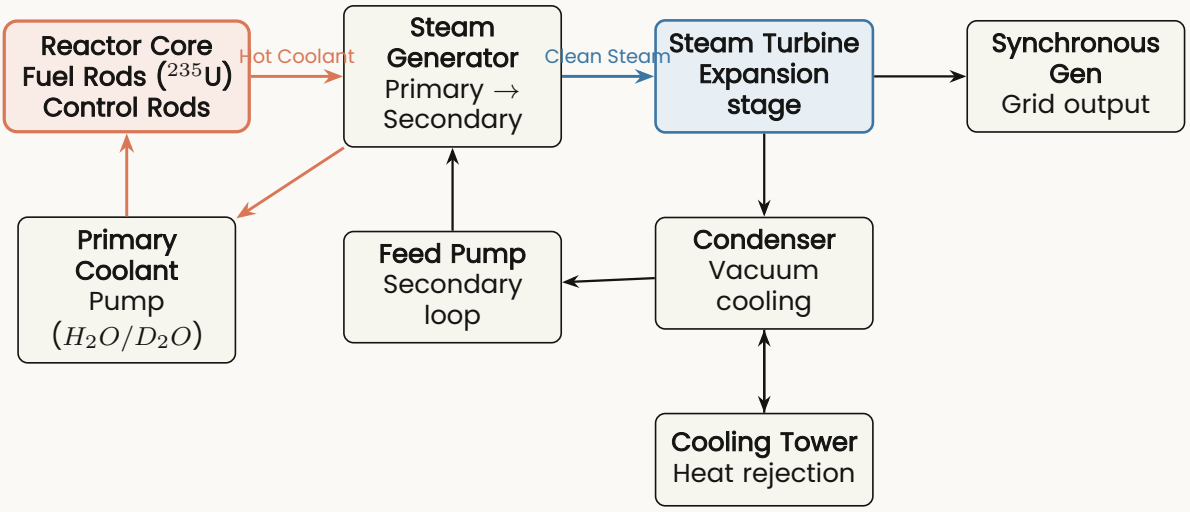
$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \longrightarrow {}^{141}_{56}\mathrm{Ba} + {}^{92}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}\mathrm{n} + \Delta E \quad (\approx 200\,\mathrm{MeV}) \tag{2}$$

The mass defect Δm is converted into thermal energy via $\Delta E = \Delta m \cdot c^2$. For sustained power, the effective multiplication factor is maintained at criticality ($k_{\text{eff}} = 1$).

Energy Conversion Chain in Nuclear Plants

$$\text{Nuclear Binding Energy} \xrightarrow{\text{Core Fission}} \text{Thermal Energy} \xrightarrow{\text{Steam Turbine}} \text{Mechanical Energy} \xrightarrow{\text{Generator}} \text{Electrical Energy}$$

Block Schematic of a Pressurized Water Reactor (PWR)



Major Reactor Components

1. **Fuel Rods:** Zircaloy tubes enclosing enriched ceramic UO_2 pellets ($3-5\%$ ^{235}U).
2. **Control Rods:** Neutron-absorbing materials (Boron Carbide, Cadmium, Hafnium) that regulate fission rate or trigger instant emergency shutdown (**SCRAM**).
3. **Moderator:** Slows fast fission neutrons (~ 2 MeV) to thermal energies (~ 0.025 eV) through elastic scattering (H_2O, D_2O, graphite).
4. **Coolant:** Circulates through the core under high pressure ($15-16$ MPa in PWR) to transport fission heat without boiling.
5. **Steam Generator:** Heat exchanger that isolates the radioactive primary loop from the non-radioactive secondary steam loop.
6. **Containment Structure:** A massive prestressed concrete and steel dome ($1-2$ m thick) designed to contain radioactive releases during severe accidents.

Fuel Energy Density & Submarine Endurance

A single uranium fuel pellet (≈ 7 g) produces energy equivalent to **nearly 1 metric tonne of coal**, 560 litres of oil, or 480 m^3 of gas. An individual's entire lifetime electricity fuel requirement fits in a standard **330 mL beverage can**. Furthermore, nuclear submarines can stay submerged for months without refueling because a single reactor operates for years on one fuel load.

Prominent Stations & Nuclear Safety Case Studies

- **Kudankulam Nuclear Power Plant (Tamil Nadu):** India's largest nuclear plant (2×1000 MW VVER-1000 PWRs, expanding to $6,000$ MW).
- **Kashiwazaki-Kariwa (Japan):** World's largest nuclear facility by installed capacity ($7,965$ MW spanning over 4.2 km^2, equal to 600 football fields).
- **Chernobyl Disaster (April 26, 1986, Ukraine/USSR):** Unauthorized safety test on an RBMK reactor with a positive void coefficient and flawed graphite-tipped control rods caused runaway prompt supercriticality, steam explosions, and destruction of the reactor building. Over 100,000 people were evacuated, leaving a 30 km permanent exclusion zone. *Lesson: Safe reactor design, proper training, strict procedures, and elimination of positive reactivity feedback.*
- **Fukushima Daiichi Disaster (March 11, 2011, Japan):** A magnitude 9.0 earthquake and 14m tsunami overtopped the 5.7m seawall, submerging emergency diesel generators and causing complete **Station Blackout (SBO)**. Loss of cooling caused core meltdowns and hydrogen

explosions across three units ($150,000$ evacuated). *Lesson: Plants must withstand extreme natural disasters, retain flood-proof power diversity, and deploy passive cooling systems.*

Advantages & Disadvantages of Nuclear Power

Advantages: Extremely high fuel energy density; near-zero operational greenhouse emissions; high capacity factors ($> 90\%$); long plant operating life ($40 - 60 +$ years).
Disadvantages: Very high capital cost and lengthy licensing; long-lived High-Level Waste (HLW) management; public risk perception; complex decommissioning.

1.5 Choosing the Right Power Generation Method

Generation Selection Matrix

- **Thermal Power Plants:** Best for reliable continuous base-load electricity where coal or gas is readily available.
- **Nuclear Power Plants:** Best for large-scale, low-carbon electricity generation with steady baseload output.
- **Hydroelectric Power Plants:** Best for mountainous regions with abundant rivers, excellent for peak-load and frequency support.
- **Solar Power Plants:** Best for high solar irradiance regions and distributed rooftop electricity generation.
- **Wind Power Plants:** Best for coastal areas, mountain passes, and open plains with strong, consistent wind flows.

2 Variable Renewable Energy Systems

Syllabus Requirement: Solar and wind – (block schematic details, environmental factors, regulations, advantages, disadvantages)

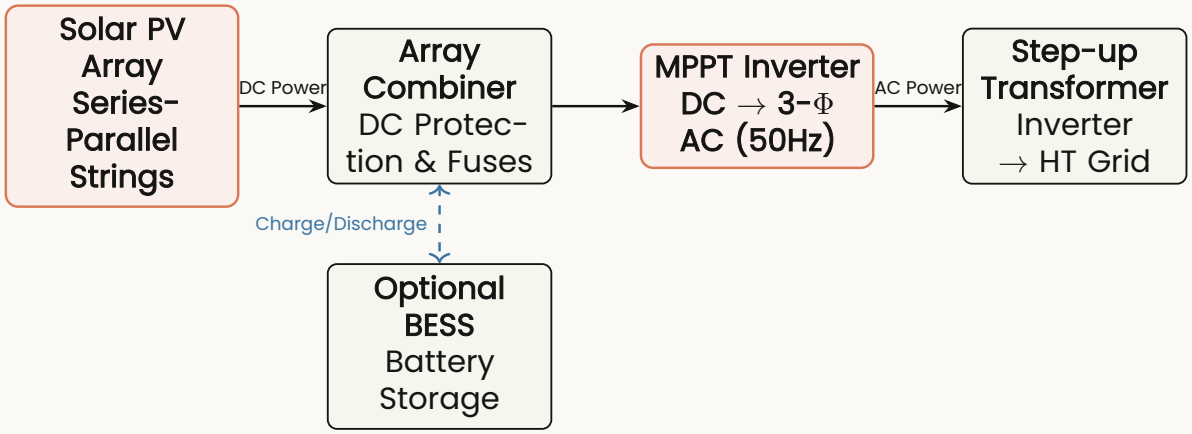
2.1 Solar Energy Systems

Solar energy is captured via two primary routes: **Solar Photovoltaics (PV)** and **Solar Thermal (Concentrated Solar Power - CSP)**.

1. Solar Photovoltaic (PV) Generation

Solar PV systems convert sunlight directly into DC electricity via the solid-state **photovoltaic effect** in semiconductor p-n junctions.

Photovoltaic Effect: Incident photons with energy $h\nu \geq E_g$ create electron-hole pairs in the depletion region. The internal electric field sweeps electrons to the n-side and holes to the p-side, producing a DC voltage ($0.5 - 0.7$ V per silicon cell).



- **PV Modules & Tracking:** Cells connected in series/parallel. Trackers (single/dual-axis) follow the sun to increase generation by $15 - 25\%$.
- **MPPT Inverter:** Converts DC to 3-phase AC while tracking maximum power points across changing solar irradiance.
- **Floating Solar PV (FPV):** Panels installed on water bodies (e.g., *NTPC Harippad / Kayamkulam 92 MW Floating Solar Project*), saving land, reducing water evaporation, and boosting module efficiency by $5 - 10\%$ through natural water cooling.
- **Famous Solar Parks:** *Bhadla Solar Park (Rajasthan, 2,245 MW)* — situated in the Thar Desert where ambient temperatures exceed $45°$C; *Golmud Solar Park (Qinghai, China, >2,800 MW)* — massive high-altitude solar array visible from space.

2. Solar Thermal (Concentrated Solar Power - CSP)

Uses mirrors (parabolic troughs or central towers) to focus solar irradiance onto a receiver, heating molten salt ($400 - 600°$C) to drive a steam Rankine turbine. **Key Advantage:** Stored thermal energy enables continuous dispatch **for 6 to 16 hours after sunset** (e.g., *Noor Ouarzazate, Morocco, 580 MW*).

Comparison: Solar PV vs. Solar Thermal (CSP)

Feature	Solar Photovoltaic (PV)	Solar Thermal (CSP)
Conversion	Direct: Photons → DC Electricity	Thermodynamic: Photons → Heat → Steam → AC
Storage	Electrochemical batteries (BESS)	Molten salt thermal storage (TES)
Water Usage	Minimal (cleaning only)	High (condenser cooling)
Dispatchability	Intermittent without BESS	Dispatchable into night peak via TES

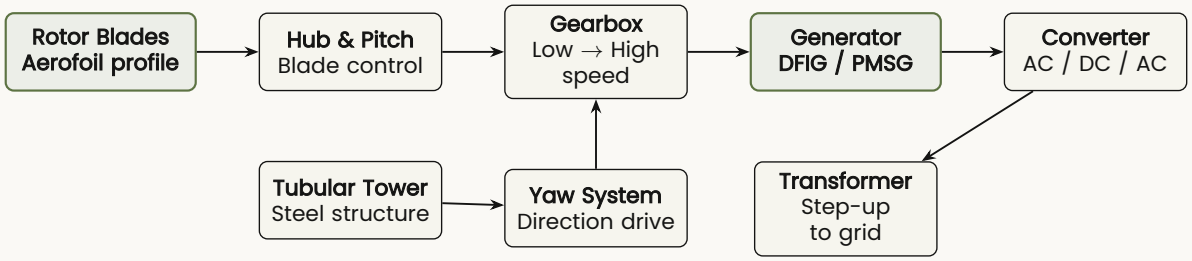
2.2 Wind Energy Systems

Working Principle & Aerodynamics

Wind turbines convert atmospheric kinetic energy into mechanical shaft power via aerodynamic lift on aerofoil blades:

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3 \quad [\text{Watts}], \qquad P_{\text{mech}} = \frac{1}{2}\rho A v^3 C_p(\lambda, \beta) \tag{3}$$

where maximum theoretical aerodynamic efficiency is governed by the **Betz Limit** ($C_{p,\max} = 16/27 \approx 59.3\%$).



- **Pitch Control:** Adjusts blade angle along its axis to optimize aerodynamic lift or feather blades in high winds.
- **Gearbox & Generator:** Steps up low rotor speed ($10-20$ RPM) to generator speed (1500 RPM); uses DFIG or direct-drive PMSG.
- **Yaw Mechanism:** Keeps the nacelle facing directly into the wind based on wind-vane signals.
- **Onshore vs Offshore:** Offshore captures stronger, steadier marine winds yielding **capacity factors** $> 50\%$.
- **Notable Projects:** *Muppandal Wind Farm (Tamil Nadu, 1500 MW)*, capturing Western Ghats mountain pass winds; *Gansu Wind Farm (China, 20,000 MW planned)*.
- **Environmental Impacts:** Low-frequency acoustic noise, shadow flicker, landscape aesthetics, and bird/bat collision risks.

2.3 Renewable Energy Policies, Regulations & Future Horizons

1. **National Solar Mission (JNNSM 2010, India):** Catalyzed competitive bidding, driving solar tariffs from over Rs. 12/kWh (2010) down below Rs. 2.50/kWh.
2. **Renewable Purchase Obligation (RPO):** Legal mandate under the Electricity Act 2003 requiring utilities and open-access consumers to purchase a designated percentage of renewable power (Solar, Non-solar, HPO).

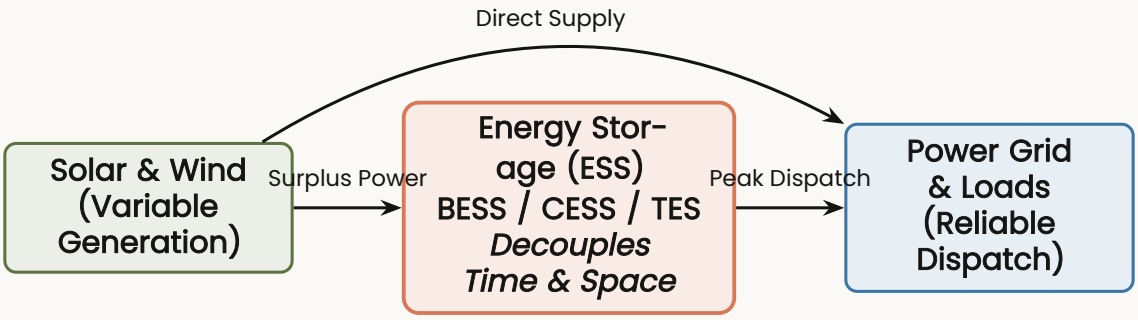
3. **PM Surya Ghar: Muft Bijli Yojana (2024):** Subsidies for 1 crore households for up to 3 kW rooftop solar, providing up to 300 free units/month.
4. **National Wind-Solar Hybrid Policy (2018):** Co-locates wind and solar at shared evacuation substations to optimize transmission utilization.
5. **India's 500 GW Target (2030):** National commitment to install 500 GW non-fossil capacity by 2030.
6. **Global Policies:** Renewable Portfolio Standards (RPS), Feed-in Tariffs (FiT), Reverse Auctions, Net/Gross Metering, Carbon Pricing / Emissions Trading Systems (ETS).
7. **Supply Chain Ethics & Agrivoltaics:** Managing environmental footprints from mining **lithium, cobalt, and rare earth elements** in fragile ecosystems; adopting **agrivoltaics** to allow dual-use farming beneath solar panels.
8. **Future Horizons: Green Hydrogen** via electrolysis for seasonal multi-day storage; **Nuclear Fusion** (ITER project, 2040s horizon) for baseload carbon-free energy.

3 Energy Storage Systems (ESS) as Alternative Sources

Syllabus Requirement: Energy storage systems as alternative energy sources – BESS, CESS, thermal SS

3.1 Need for Energy Storage in Modern Grids

Energy storage addresses renewable intermittency, mitigates steep evening ramping (*the Duck Curve*), provides synthetic inertia, captures curtailed energy, and supports frequency regulation.



3.2 Battery Energy Storage Systems (BESS)

- **Chemistries:** Lithium Iron Phosphate ($LiFePO_4$ / LFP) for thermal safety and cycle life; Nickel Manganese Cobalt (NMC) for energy density; Flow Batteries (Vanadium Redox) for long-duration bulk storage; emerging Solid-State batteries.
- **Performance:** Fast sub-second response (< 100 ms) for frequency regulation; high round-trip efficiency ($85 - 95\%$); $1 - 4$ hour discharge duration; requires Battery Management Systems (BMS) with liquid cooling to prevent thermal runaway.
- **Benchmark:** *Hornsdale Power Reserve (Tesla Megapack, 150 MW / 193.5 MWh, South Australia).*

3.3 Compressed Air Energy Storage (CESS / CAES)

- **Mechanism:** Off-peak power compresses air ($40 - 80$ bar) into underground sealed salt caverns, depleted aquifers, or rock mines; peak demand releases air through preheaters and expander turbines driving generators.
- **Features:** Multi-hour to multi-day bulk capacity (hundreds of MWh); long life (> 30 years); efficiency $40 - 70\%$; tied to geological formations.

3.4 Thermal Energy Storage (TES / Thermal SS)

- **Media:** Molten salts ($290 - 565°C$), pressurized hot water, Phase Change Materials (PCMs), chilled water/ice for HVAC peak shaving.
- **Applications:** CSP evening generation, industrial process heat, and district heating/cooling.

3.5 Comprehensive Comparison Matrix of Storage Technologies

Technology	Efficiency	Response	Duration	Primary Grid Services
BESS (Li-ion)	85 – 95%	Milliseconds	15 min - 4 hrs	Frequency regulation, synthetic inertia, fast peak shaving
CESS (CAES)	40 – 70%	5 - 15 min	4 - 24+ hrs	Bulk energy shifting, seasonal balancing
Thermal SS (TES)	50 – 80%	Minutes	4 - 16 hrs	CSP night dispatch, HVAC load shifting
Pumped Hydro (PSH)	70 – 85%	1 - 3 min	6 - 20+ hrs	Bulk baseload storage, black start

4 Load Curves, Station Economics & Mathematical Formulations

Syllabus Requirement: Load curve – Load duration curve, Load factor, diversity factor, demand factor, Plant capacity factor, plant use factor - Numerical Problems.

4.1 Load Dynamics and Operational Challenges

The Invisible Challenge: Why Power Cannot Wait

Unlike gas or water, bulk electrical energy cannot be warehoused directly on transmission lines. Generation must match consumption instantaneously across shifting human behavior.

When Variables Collide

- **Equipment Wear:** Frequent load cycling induces thermal fatigue in turbine blades and boiler tubes, accelerating maintenance schedules.
- **Fuel Efficiency:** Thermal plants operated below rated capacity suffer a $15 - 25\%$ **part-load heat rate penalty**.
- **System Responsiveness:** Fast load changes demand unloaded spinning reserves, imposing an efficiency penalty across the fleet.

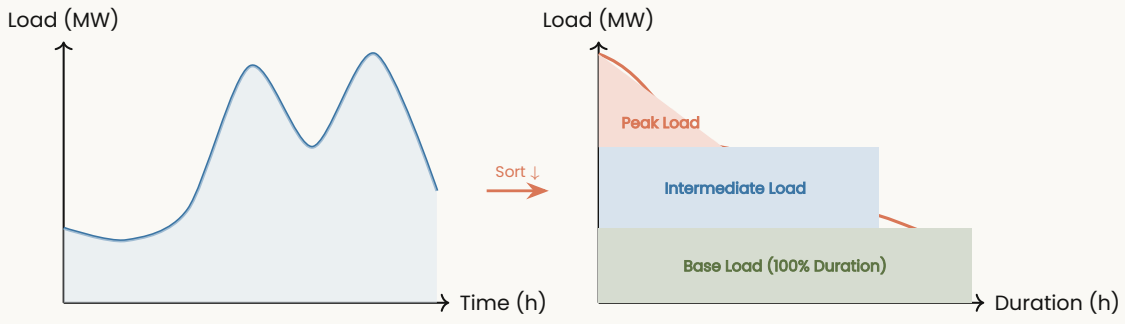
The Future of Load Management

- **Smart Grids:** Real-time bidirectional communication enabling automated **Demand-Side Management (DSM)** to flatten load curves.
- **AI-Driven Forecasting:** Machine learning models predict demand surges hours in advance using weather and historical data.
- **Dynamic Grid Intelligence:** Self-optimizing grid systems balancing cost, carbon emissions, and reliability.

4.2 The Load Curve and Load Duration Curve

Load Curve vs. Load Duration Curve: **Load Curve:** Chronological plot of power demand (kW/MW) vs time (hours). Area under curve = Total Energy (E).
Load Duration Curve (LDC): Chronological load rearranged in **descending order of magnitude**.

$$E = \int_0^T P(t)\,dt = \sum_{i=1}^{n} P_i \cdot \Delta t_i \quad \text{[kWh or MWh]}, \qquad L_{\text{avg}} = \frac{E}{T} \quad \text{[kW or MW]} \tag{4}$$



(a) Daily Load Curve (b) Load Duration Curve

4.3 Mathematical Definitions of Economic Factors

1. **Connected Load:** Continuous sum of continuous ratings of all connected appliances.

2. **Maximum Demand (MD):** Greatest load drawn during a specified 15/30-minute interval.
3. **Demand Factor (DF):**

$$DF = \frac{\text{Maximum Demand}}{\text{Connected Load}} \leq 1 \tag{5}$$

Typical values: Domestic $0.4 - 0.6$*, Commercial* $0.5 - 0.8$*, Industrial* $0.7 - 0.95$*.*

4. **Load Factor (LF):**

$$LF = \frac{\text{Average Load}}{\text{Maximum Demand}} = \frac{\text{Energy Generated in time } T}{MD \times T} \leq 1 \tag{6}$$

5. **Diversity Factor ($DivF$):**

$$DivF = \frac{\sum_{k=1}^{n} MD_k}{\text{Coincident System Maximum Demand}} \geq 1 \tag{7}$$

6. **Plant Capacity Factor (PCF):**

$$PCF = \frac{\text{Actual Energy Generated}}{\text{Installed Plant Capacity} \times T} = \frac{L_{\text{avg}}}{\text{Installed Capacity}} \leq 1 \tag{8}$$

$$\text{Reserve Capacity} = \text{Installed Plant Capacity} - \text{Maximum Demand} \tag{9}$$

$$PCF = LF \times \left(\frac{MD}{\text{Installed Capacity}}\right) = LF \times \text{Utilization Factor} \tag{10}$$

7. **Plant Use Factor (PUF):**

$$PUF = \frac{\text{Actual Energy Generated}}{\text{Installed Capacity} \times T_{\text{operating}}} \geq PCF \tag{11}$$

8. **Utilization Factor (UF):** $UF = \frac{\text{Maximum Demand}}{\text{Installed Capacity}} \leq 1$.

The Power Engineer's Mandate

A **1% improvement in load factor** across a $1,000$ MW generation fleet represents millions of dollars in avoided capital and fuel expenditure annually.

4.4 Fully Worked-Out Numerical Problems

Problem 1: Residential Distribution Feeder & Capacity Sizing

A housing township has 3 groups: Group A (50 homes, connected load $2,500$ kW, $DF = 0.5$), Group B (30 shops, connected load $1,800$ kW, $DF = 0.6$), Group C (20 community facilities, connected load $1,200$ kW, $DF = 0.55$). Diversity factor = 1.40. Find individual max demands, sum, system max demand, transformer rating, and capacity saving.

Step-by-Step Solution

1. Individual Demands: $MD_A = 2500 \times 0.5 = 1,250$ kW, $MD_B = 1800 \times 0.6 = 1,080$ kW, $MD_C = 1200 \times 0.55 = 660$ kW.
2. Sum of Individual Peaks: $\sum MD_i = 1250 + 1080 + 660 = \mathbf{2,990}$ kW.
3. System Coincident Peak: $MD_{\text{system}} = \frac{2990}{1.40} = \mathbf{2,135.71}$ kW $\approx \mathbf{2,136}$ kW.
4. Transformer Rating Required = **2,136 kW**.
5. Capacity Saving: $\frac{5500 - 2135.71}{5500} \times 100\% = \mathbf{61.17\%}$.

Problem 2: Daily Step-Load Schedule Analysis

A 100 MW station has the daily schedule: 00 – 06h (40 MW), 06 – 12h (80 MW), 12 – 18h (100 MW), 18 – 24h (60 MW). Find daily energy, annual energy, average load, load factor, and plant capacity factor.

Step-by-Step Solution

1. Daily Energy: $E_{\text{daily}} = (40 \times 6) + (80 \times 6) + (100 \times 6) + (60 \times 6) = 240 + 480 + 600 + 360 = \mathbf{1,680}$ MWh.
2. Annual Energy: $E_{\text{annual}} = 1680 \times 365 = \mathbf{613,200}$ MWh.
3. Average Load: $L_{\text{avg}} = \frac{1680}{24} = \mathbf{70}$ MW.
4. Load Factor: $LF = \frac{70}{100} = \mathbf{0.70} = \mathbf{70\%}$.
5. Plant Capacity Factor: $PCF = \frac{70}{100} = \mathbf{70\%}$.

Problem 3: Comparative Economic Evaluation of Two Power Stations

Plant A (100 MW cap, 80 MW peak) and Plant B (150 MW cap, 100 MW peak) each generate 500, 000 MWh annually. Compare average load, load factor, capacity factor, reserve capacity, and economics.

Step-by-Step Solution

1. Average Load: $L_{\text{avg}} = \frac{500,000}{8760} = \mathbf{57.08}$ MW for both plants.
2. Load Factor: $LF_A = \frac{57.08}{80} = \mathbf{71.35\%}$; $LF_B = \frac{57.08}{100} = \mathbf{57.08\%}$.
3. Capacity Factor: $PCF_A = \frac{57.08}{100} = \mathbf{57.08\%}$; $PCF_B = \frac{57.08}{150} = \mathbf{38.05\%}$.
4. Reserve: $\text{Res}_A = 100 - 80 = \mathbf{20}$ MW; $\text{Res}_B = 150 - 100 = \mathbf{50}$ MW.
5. **Economic Verdict:** Plant A generates the same energy from lower capital capacity, achieving much higher load factor and lower unit generation cost.

Problem 4: Load Factor Calculation

Maximum demand = 80 MW, Daily energy = 1200 MWh. Find Average Load and Load Factor.

Step-by-Step Solution

$L_{\text{avg}} = \frac{1200}{24} = \mathbf{50}$ MW, $\quad LF = \frac{50}{80} = \mathbf{0.625} = \mathbf{62.5\%}$.

Problem 5: Demand Factor Calculation

Connected Load = 500 kW, Maximum Demand = 350 kW. Find Demand Factor.

Step-by-Step Solution

$DF = \frac{350}{500} = \mathbf{0.70} = \mathbf{70\%}$.

Problem 6: Diversity Factor Calculation

Consumer demands: $40, 60, 80, 100$ kW. System coincident peak = 200 kW. Find Diversity Factor.

Step-by-Step Solution

$\sum MD_i = 40 + 60 + 80 + 100 = 280$ kW, $\quad DivF = \frac{280}{200} = \mathbf{1.40}$.

Problem 7: Plant Capacity Factor Calculation

Capacity = 100 MW, Daily Energy = 1440 MWh. Find Plant Capacity Factor.

Step-by-Step Solution

$PCF = \frac{1440}{100 \times 24} = \frac{1440}{2400} = \mathbf{0.60} = \mathbf{60\%}$.

Problem 8: Plant Use Factor with Operating Outages

Capacity = 120 MW, Operating time = 16 hours/day, Daily Energy = 1440 MWh. Find Plant Use Factor and Capacity Factor.

Step-by-Step Solution

$PUF = \frac{1440}{120 \times 16} = \frac{1440}{1920} = \mathbf{0.75} = \mathbf{75\%}$, $\quad PCF = \frac{1440}{120 \times 24} = \frac{1440}{2880} = \mathbf{50\%}$.

Problem 9: Multi-Parameter Station Analysis

Installed Capacity = 150 MW, Maximum Demand = 120 MW, Connected Load = 180 MW, Daily Energy = 2160 MWh. Determine: (a) L_{avg}, (b) LF, (c) DF, (d) PCF, (e) Reserve Capacity.

Step-by-Step Solution

1. $L_{\text{avg}} = \frac{2160}{24} = \mathbf{90}$ MW.
2. $LF = \frac{90}{120} = \mathbf{0.75} = \mathbf{75\%}$.
3. $DF = \frac{120}{180} = \mathbf{0.667} = \mathbf{66.7\%}$.
4. $PCF = \frac{2160}{150 \times 24} = \mathbf{0.60} = \mathbf{60\%}$.
5. Reserve $= 150 - 120 = \mathbf{30}$ MW.

Problem 10: Complete Operational Analysis

Capacity = 200 MW, Operating Hours = 18 h, Daily Energy = 2520 MWh, Connected Load = 240 MW, $MD = 180$ MW. Determine L_{avg}, LF, DF, PCF, PUF, and UF.

Step-by-Step Solution

1. $L_{\text{avg}} = \frac{2520}{24} = \mathbf{105}$ MW.
2. $LF = \frac{105}{180} = \mathbf{0.5833} = \mathbf{58.33\%}$.

3. $DF = \frac{180}{240} = \mathbf{0.75} = \mathbf{75\%}$.
4. $PCF = \frac{2520}{200\times24} = \frac{2520}{4800} = \mathbf{52.5\%}$.
5. $PUF = \frac{2520}{200\times18} = \frac{2520}{3600} = \mathbf{70\%}$.
6. $UF = \frac{180}{200} = \mathbf{0.90} = \mathbf{90\%}$.

Problem 11: Linear Load Duration Curve Integration

An annual load duration curve slopes linearly from 120 MW to 30 MW over 8,760 hours. Find total energy, average load, load factor, and base/peak energy if a 40 MW base plant is used.

Step-by-Step Solution

$$E_{\text{total}} = \left(\frac{120+30}{2}\right) \times 8760 = 75 \times 8760 = \mathbf{657,000}\,\text{MWh}$$

$$L_{\text{avg}} = \frac{657,000}{8760} = \mathbf{75}\,\text{MW}, \qquad LF = \frac{75}{120} = \mathbf{0.625} = \mathbf{62.5\%}$$

$$E_{\text{base}} = 40 \times 8760 = \mathbf{350,400}\,\text{MWh} \quad (53.33\%)$$

$$E_{\text{peak}} = 657,000 - 350,400 = \mathbf{306,600}\,\text{MWh} \quad (46.67\%)$$

Problem 12: Generation Tariff Economics

A 50 MW station operates at annual load factor 60%. Capital cost = Rs. 5 crore/MW, annual fixed charges = 12%, fuel/operating cost = Rs. 2.20/kWh. Calculate total annual cost and cost per kWh generated.

Step-by-Step Solution

$$L_{\text{avg}} = 50,000\,\text{kW} \times 0.60 = 30,000\,\text{kW}$$

$$E_{\text{annual}} = 30,000 \times 8760 = 262.8 \times 10^6\,\text{kWh}$$

$$\text{Fixed Cost} = 12\% \times (50 \times \text{Rs. } 5 \times 10^7) = \text{Rs. } 30 \times 10^7 = \text{Rs. } 300 \times 10^6$$

$$\text{Variable Cost} = 262.8 \times 10^6 \times \text{Rs. } 2.20 = \text{Rs. } 578.16 \times 10^6$$

$$\text{Total Cost} = \text{Rs. } 878.16 \times 10^6$$

$$\text{Cost per kWh} = \frac{878.16 \times 10^6}{262.8 \times 10^6} = \text{Rs. } \mathbf{3.34}/\text{kWh} \quad (\text{Fixed: Rs. } 1.14, \text{ Variable: Rs. } 2.20)$$

5 Quick Revision Sheet & Model Examination Questions

5.1 Master Formula and Definition Summary Table

Factor / Metric	Formula	Typical Value	Physical Significance
Demand Factor (DF)	$\frac{\text{Maximum Demand}}{\text{Connected Load}}$	≤ 1 $(0.4 - 0.8)$	Non-simultaneous load operation; sizes feeder rating
Load Factor (LF)	$\frac{\text{Average Load}}{\text{Maximum Demand}} = \frac{\text{Energy}}{MD \times T}$	≤ 1 $(0.5 - 0.85)$	Steady plant utilization; higher LF cuts unit generation cost
Diversity Factor ($DivF$)	$\frac{\sum \text{Individual Peaks}}{\text{System Peak Demand}}$	≥ 1 $(1.2 - 1.6)$	Staggered consumer peaks; reduces required station size
Capacity Factor (PCF)	$\frac{\text{Actual Energy}}{\text{Installed Capacity} \times T} = \frac{L_{\text{avg}}}{\text{Cap}}$	≤ 1 $(0.4 - 0.8)$	Capital asset utilization index
Reserve Capacity	Installed Capacity $- MD$	≥ 0 MW	Safeguards against outages and peak surges
Plant Use Factor (PUF)	$\frac{\text{Actual Energy}}{\text{Installed Capacity} \times T_{\text{op}}}$	$\geq PCF$	Operational efficiency during active run-time
Utilization Factor (UF)	$\frac{\text{Maximum Demand}}{\text{Installed Capacity}}$	≤ 1 $(0.7 - 0.95)$	Fraction of station capacity demanded at peak

5.2 Master Comparison Matrix of All Power Generation Sources

Feature	Hydroelectric	Thermal (Coal)	Nuclear	Solar PV	Wind Energy
Primary Source	Gravitational potential	Chemical (coal/gas)	Nuclear fission (^{235}U)	Solar photons	Kinetic wind flow
Cycle / Mode	Hydraulic turbine	Rankine / Brayton	Fission $\rightarrow$ Rankine	Solid-state PV effect	Aerodynamic lift
Carbon Emissions	Negligible	Very High (CO_2)	Negligible	Negligible (operation)	Negligible
Capital Cost	Very High	Moderate	Extremely High	Low to Moderate	Moderate
Operating Cost	Minimal (no fuel)	High (fuel recurring)	Low to Moderate	Minimal	Minimal
Start-up Time	Fast ($2 - 5$ min)	Slow ($4 - 8$ hrs)	Very Slow (Days)	Instantaneous	Fast
Capacity Factor	$30 - 60\%$ (seasonal)	$60 - 80\%$	$> 90\%$	$18 - 25\%$	$25 - 45\%$
Grid Role	Peaking / Baseload	Baseload	Baseload	Variable / Peaking	Variable
Key Vulnerability	Monsoon dependency	Greenhouse emissions	Waste & Meltdown	Night intermittency	Wind intermittency

5.3 Model Examination Questions (KTU S5 Pattern)

University Examination Pattern (PCEOT503 Scheme)

The End Semester Examination consists of **Part A** (8 compulsory questions of 3 marks each, 2 from each module) and **Part B** (Module choice questions carrying 9 marks each).

Part A: Short Answer Questions (3 Marks Each)

1. **Q1:** Define *Load Factor* and *Diversity Factor*. Explain why a high diversity factor is economically desirable in power systems.
2. **Q2:** Differentiate between *Plant Capacity Factor* and *Plant Use Factor*. Under what operational condition are they identical?
3. **Q3:** Explain the essential function of a *Surge Tank* in a hydroelectric power installation.
4. **Q4:** State the *Photovoltaic Effect* and explain how direct current is produced in a p-n junction solar cell.
5. **Q5:** What is meant by *Station Blackout (SBO)* in nuclear plants? Mention one key safety lesson learned from the Fukushima disaster.
6. **Q6:** Why are *Battery Energy Storage Systems (BESS)* considered superior to conventional spinning reserve for fast frequency regulation?

Part B: Long Answer / Analytical Questions (9 Marks Each)

1. **Q7 (a):** Draw the neat schematic block layout of a modern coal-fired thermal power station and describe the functional role of the Economiser, Superheater, Air Preheater, and Surface Condenser. (6 Marks)
   Q7 (b): Discuss the ethical and environmental dilemmas associated with coal-fired thermal power generation. (3 Marks)
2. **Q8 (a):** Explain the working of a nuclear power reactor with a neat block diagram. Discuss the functions of the moderator, control rods, and containment structure. (5 Marks)
   Q8 (b): A generating station has a maximum demand of 50 MW, a connected load of 80 MW, and generates 180×10^6 kWh annually. The installed capacity is 60 MW. Calculate: (i) Demand Factor, (ii) Load Factor, (iii) Plant Capacity Factor, and (iv) Reserve Capacity. (4 Marks)
3. **Q9 (a):** With the aid of a neat block diagram, explain the components of a horizontal-axis wind turbine generator system. Distinguish between pitch control and yaw control. (5 Marks)
   Q9 (b): Explain the role of Energy Storage Systems (BESS, CESS, and Thermal SS) in enabling high penetration of variable renewable energy on the grid. (4 Marks)

— End of Module 1 Study Notes —